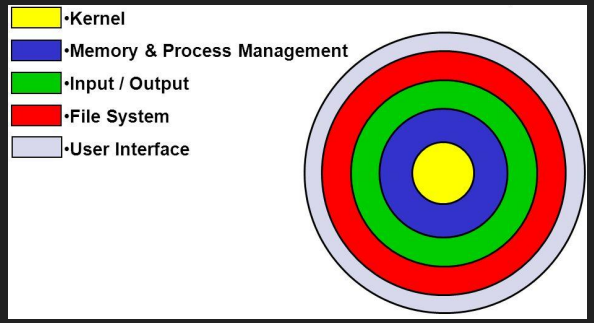
Task 1 - Explain the purpose of operating systems

Leandro Baroca

Functions of Windows Operating Systems

Kernel



In an operating system, the kernel is basically the core of the operating system, and uses interprocess communication and system calls. What I mean by this is that it acts like a bridge between applications and the data processing performed at the hardware level.

If an operating system is being loaded into the memory, the kernel loads first and remains in memory until the operating system is shut down again.

The kernel is responsible for many low-level tasks, such as: disk management, task management, memory management, process management for application execution, device management through use of device drivers and the system call control, which is essential for the executions of kernel services.

User Interface

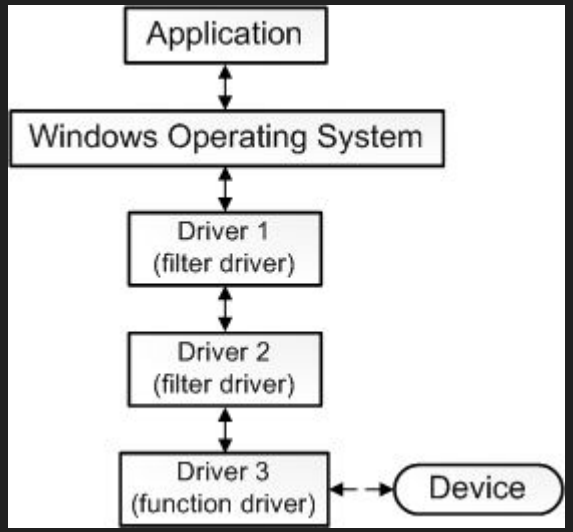


The user interface, or “UI”, is the software which a person can control such as a software application or a hardware device. Software programs have a GUI, or graphical user interface. What I mean by this is that the program has graphical controls, which gives the user the ability to use a mouse or keyboard.

It is important that the user interface gives a user friendly experience to the user because it gives the user a much more natural and smooth way of using the software or hardware. A UI which isn't user friendly or overly complicated will be seen as a badly made UI.

A UI does not only apply to software, hardware devices can also have a user interface, hardware UI is a lot simpler than the more complex software interface. An example of a hardware UI is a remote control for a TV.

Device Drivers

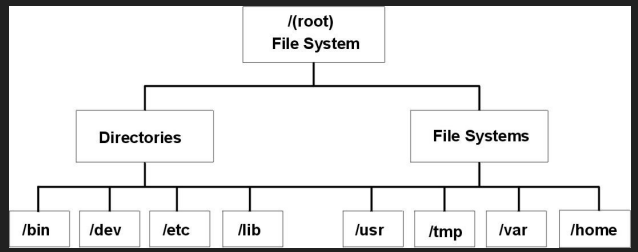


The device driver is basically a small program that enables a computer and a device to communicate with each other. Operating systems on a computer usually comes with preinstalled device drivers. However, the device driver are compatible with the the popular devices on the time it was installed or updated.

Device drivers don't have an automatic update, so that means that the older or newer versions of drivers have to be manually installed by the user. Some Windows programs are virtual device drivers, which is when the programs interface with the Windows Virtual Machine Manager.

In a Windows operating system, a device driver file usually has a file name suffix of DLL or EXE. A virtual device driver usually has the suffix of VXD.

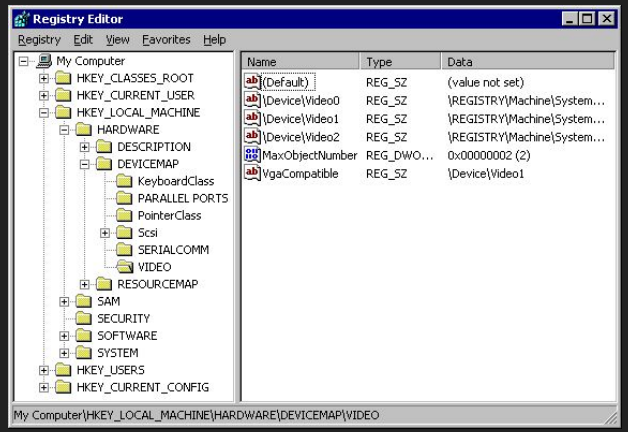
File System Management



The file system management is a system of an operating system or a program that manages the data files within a computer system. The system is designed to manage individual or group files such as special office documents and records, but the system has very limited capabilities. What I mean by this is that the system may contain simple features such as: Assigning queued document numbers for processing, owner and process mapping to track various stages of processing, report generation, notes, status, create, modify, copy, delete and other file operations.

File management should not be confused with a file system, which manages all types of data and files in an OS. Also with database management system (DBMS), which has a relational database capability and includes a programming language for data manipulation

Registry



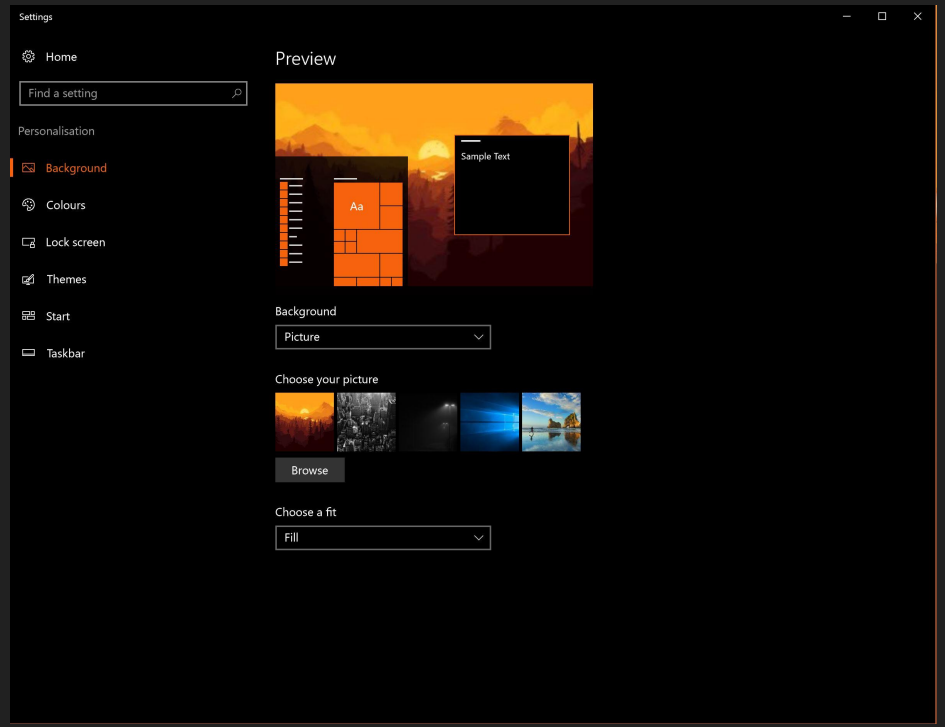
The registry on the Windows OS is used to store much of the information and settings for software programs, hardware devices, user preferences, operating system configurations and much more.

For example, if a new program is installed, new sets of instructions and file references can be added into the registry in a specific location for the program. It is thought that the registry is a bit like the DNA for the Windows operating system.

Of course, the Windows applications rely on Windows Registry, but not all applications necessarily need to utilize the Windows Registry. Some programs store their configurations within XML files instead of the registry. Other applications are entirely portable and store their data in an executable file.

Features of Windows 10 Operating System

Customizing the User Interface (Desktop, etc.)

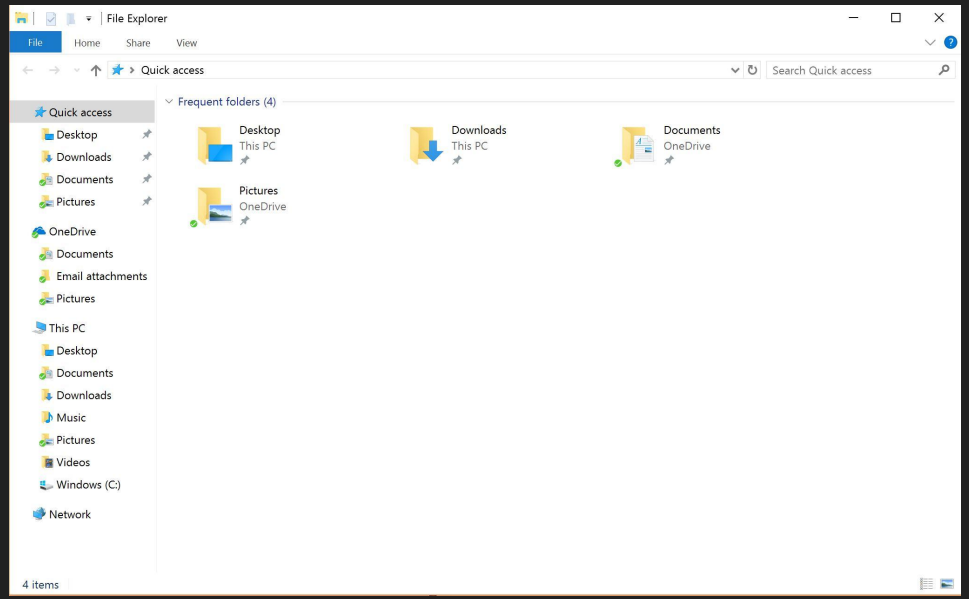


With windows 10, you can heavily customize the user interface better than the previous Windows OS. This can give you the ability to personalise it to your taste.

Windows 10 includes many personalization settings that let you change your desktop background, windows colours, lock screen background, and more.

To access this screen, you need to go to Personalization settings which is available at Settings > Personalization.

File Management

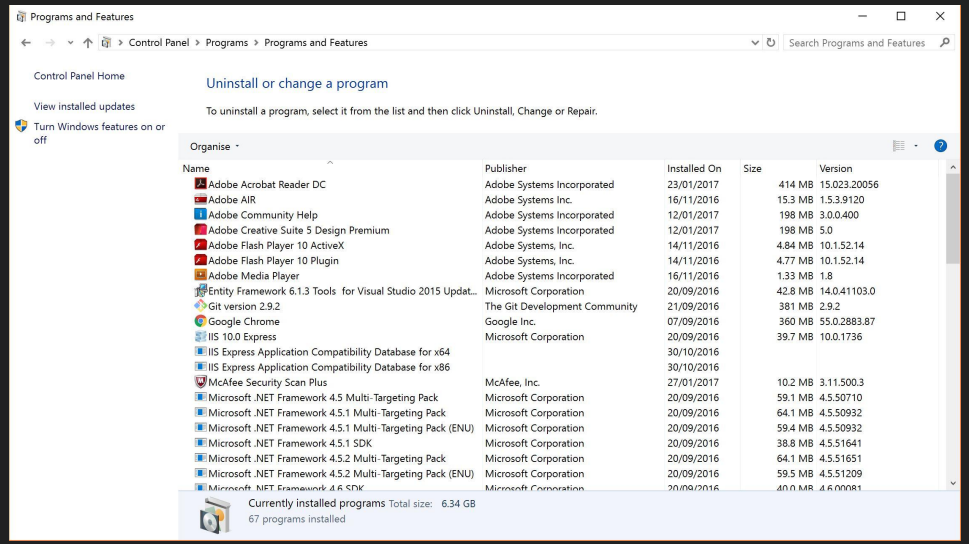


A critical part of Windows in general, is that you can organize files stored in the hard disk by using a File System that manages how files are stored and accessed on the computer.

To access File Explorer, you click Start > FIle explorer. When the File Explorer is launched, you get into the Quick access window, which pins the most frequently accessed folders.

You can use FIle Explorer for a variety of tasks. In addition to management and organization of files and folders, it’s also used to view and manage the resources of your computer such as internal storage, attached storage and optical drives.

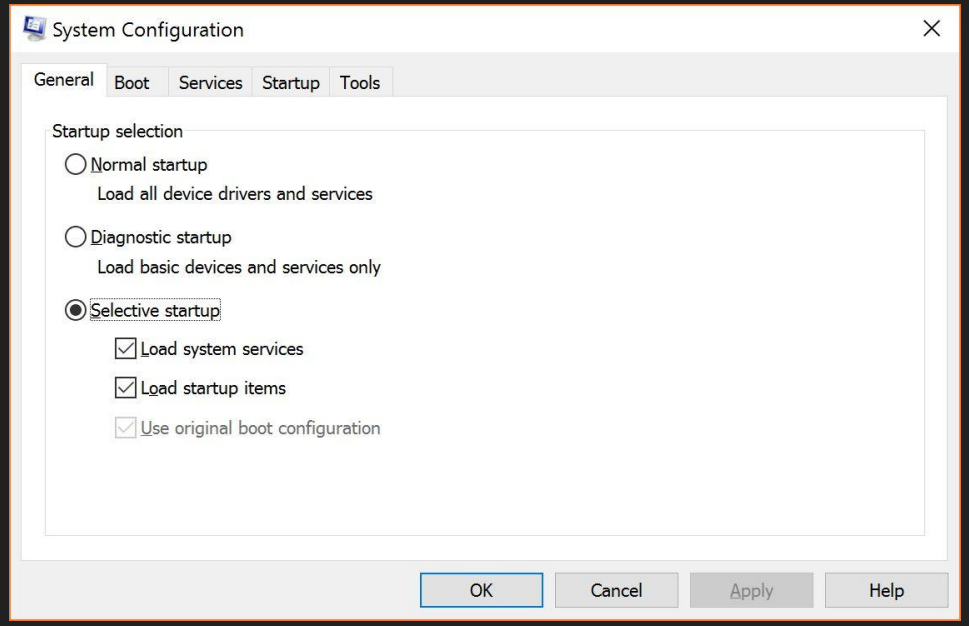
Application Management



Windows 10 offers multiple ways to access the applications. There are three ways in which you can manage and remove applications: Start Menu, Settings Window and from the Control Panel.

To open it from the start menu, you open the start menu and click on All Apps, and you will see a list of all the applications in alphabetical order. To open it from the Settings window, you search “settings” on the search bar and click on settings, click on system and choose Apps & features. To open it from the control panel, you type in “control panel” in the search bar, then you choose Programs and Features.

System Management Utilities

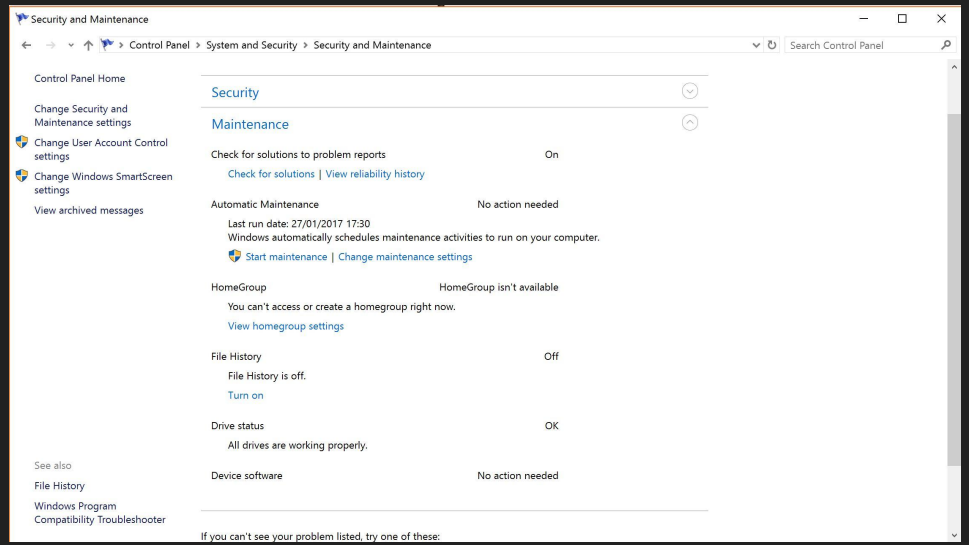


MSCONFIG is a utility built into WIndows to help with management and diagnostics of the Windows PC. By going on start > run window and typing “msconfig” a window will come up with the title “System Configuration”.

Once you're in the window, you can view all the background services that start when you boot your computer. You can also view all the programs that automatically start when you boot your computer.

Once you get inside these tabs, being able to turn items on and off will allow you to manage your memory usage and boot up times.

Windows Maintenance Tasks



Windows always had in-built tools that help customers to take care of their PC. Windows 10 has improved with the tools to make it a lot easier to take care of the PC. The automatic maintenance in WIndows 10 is the maintenance task on a windows 10 PC.

To open it you search for its settings via search box/cortana or through the control panel. Once open, you can set the time when the automatic maintenance runs its full diagnostic and cleanup. Choose a time when your computer is on and you're not using it for best results. You can also choose to let automatic maintenance in WIndows 10 turn off your PC to runs its process